

Karthik Mothiki

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PROFESSIONAL SUMMARY

Senior Robotics Systems Engineer specialising in autonomous mobile robots, modern C++, embedded Linux and distributed robotics software. Experienced architecting production robotic platforms from hardware abstraction and firmware through AI-enabled autonomy while leading multidisciplinary engineering teams.

WORK EXPERIENCE

Ogmen Robotics

New Delhi, India

Senior Robotics Engineer

May 2026 - Present

- Lead 3 cross-functional engineering streams (mechanical, electronics/hardware, software) as technical owner for [ORo Base](#), an Intelligent Canine Wellness Station.
- Integrated software, firmware, electronics, and mechanical subsystems into a production robotics platform, driving system-level architecture and cross-functional technical decisions.
- Coordinate sprint planning, dependency management, and cross-functional execution across robotics software, electronics, and mechanical engineering teams.
- Translate technical progress into execution decisions across engineering and business teams.
- Own evolution of a modular C++17 embedded software platform spanning HAL, middleware, and edge computing across 10+ sensors/actuators.
- Architected the evolution of the platform's communication and runtime infrastructure, integrating embedded firmware, hardware abstraction, and Linux services (ZeroMQ, UART, ESP32-S3, systemd, V4L2, GStreamer) into a unified software stack.

Robot Software Engineer

Oct 2024 - May 2026

- Designed and built a Task Management and Safety Execution System implementing thread-safe priority task queues and dedicated safety-monitoring nodes for deterministic scheduling and real-time orchestration. Validated queue behavior under simulated concurrent task influx from multiple input sources, confirming correct population and ordering under load.
- Architected a modular C++17 Hardware Abstraction Layer, independent of any robotics middleware framework, unifying real-time telemetry and control across 13+ sensor and actuator modules for [ORO Buddy](#), an autonomous Pet Companion Robot.
- Owned firmware and middleware for the robot's compute module in C++ on ESP32-S3/FreeRTOS, tuning and calibrating 20+ sensors and 8+ actuators, including low-level FOC motor controller integration and closed-loop motion tuning for reliable real-time actuation.
- Implemented a processor-to-processor IPC framework using socket programming, enabling low-latency bi-directional data exchange between heterogeneous compute nodes at 1 Gbps.
- Developed a vision-guided rear-end autonomous docking pipeline for a front-drive mobile robot with uneven load distribution, improving docking success from functionally undockable using standard Nav2 Docking to 78% (78/100 trials); the solution was evaluated for patent filing before project reprioritisation.

My Engineering Buddy

Remote

Professional Freelancer

Feb 2024 - Oct 2024

- Delivered freelance robotics and control systems projects involving MATLAB, Simulink, robotics algorithms, and model-based system design for academic and industry clients.

Indian Institute of Technology Madras

Chennai, India

Project Engineer

Nov 2023 - Jan 2024

- Developed a ROS interface enabling an IGUS 5-DOF manipulator for autonomous ultrasound imaging research for prenatal care.
- Integrated MoveIt and Point Cloud Library (PCL) for ToF-camera-based surface segmentation to support robotic ultrasound path planning.

CORE SKILLS

Languages: C++17, Python, C, MATLAB

Robotics: ROS2, ROS, MoveIt, BehaviorTree.CPP, Nav2, SLAM, Navigation, Managed Nodes

Embedded: Embedded Linux, ESP32-S3, FreeRTOS, UART, POSIX, systemd, Arduino

Distributed & Cloud Systems: ZeroMQ, IPC, Socket Programming, GCP Cloud Run

AI Engineering: Google ADK, Gemini 2.0, MCP, Pydantic

Motor Control: Field-Oriented Control (FOC), ODESC V4.2 Motor Controllers

PROJECTS

Sūtradhāra - Multi-Agent Productivity Orchestrator

[Github](#)

- Architected a multi-agent AI orchestration platform using Google ADK, Gemini 2.0, and MCP with FastAPI, WebSockets, and async services for workflow automation.
- Implemented HITL approval workflows, rollback, observability, Google Workspace integration, and deployed the platform on Google Cloud Run.

CppSentry - C++ Code Analyzer Agent

[Github](#)

- Built an AI-assisted C++17 static analysis platform combining deterministic analysis with LLM reasoning to evaluate memory safety, complexity, and modern C++ compliance.
- Developed secure REST APIs, structured reporting, and containerized deployment on Google Cloud Run.

ROS Modernization of Harvest Automation HV-100 Agricultural Robot

- Modernized the ROS software stack of a commercial Harvest Automation HV-100 agricultural robot by resolving ROS Melodic end-of-life package failures, driver conflicts, sensor calibration issues, and OEM controller communication faults, restoring full robot functionality.
- Designed and deployed a Raspberry Pi-based ROS Bridge control architecture enabling teleoperation, differential-drive mobility, manipulator control, and ROS-native communication with the proprietary robot controller.

Autonomous Navigation using Mobile Manipulator

- Developed a mobile manipulator simulation using KUKA youBot, implementing autonomous navigation with dynamic obstacle avoidance in ROS and CoppeliaSim.

Flower Categorization using Deep Learning

- Built and trained an Image Classifier that predicts the top 5 classes of flowers using a pre-trained deep learning model (VGG16) with an accuracy of 78.09%
 - Developed a command-line application to train models and predict the top five flower classes, generating Matplotlib visualisations alongside ground-truth labels
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EDUCATION

Sastra Deemed to be University

B.Tech Mechatronics

Thanjavur, TN

2019 - 2023

Exchange Semester & Bachelor's Thesis

Örebro Universitet, Sweden

Jan - Jun 2023

CERTIFICATIONS

- *ASPIRE Leaders Program* *Harvard Business School*
- *Robotics Software Engineer Nanodegree* *Udacity*
- *AI Programming with Python Nanodegree* *Udacity*
- *Matlab Onramp & Simulink Onramp* *Mathworks*